EXTENSION 1

Stage 6

Combinatorics (Ext1)

A1 Working with Combinatorics (Y11)

Binomial Expansion

Teacher: Karen Howlett

Exam Equivalent Time: 18 minutes *(based on allocation of 1.5 minutes per mark)*

Blackwattle Bay

Questions

1. Combinatorics, EXT1 A1 2014 HSC 3 MC

What is the constant term in the binomial expansion of $\left(2x - \dfrac{5}{x^3}\right)^{12}$?

A. $\binom{12}{3} 2^9 5^3$

B. $\binom{12}{9} 2^3 5^9$

C. $-\binom{12}{3} 2^9 5^3$

D. $-\binom{12}{9} 2^3 5^9$

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2. Combinatorics, EXT1' A1 2016 HSC 6 MC

Let $p(x) = 1 + x + x^2 + x^3 + \ldots + x^{12}$.

What is the coefficient of x^8 in the expansion of $p(x+1)$?

A. 1

B. 495

C. 715

D. 1287

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3. Combinatorics, EXT1 A1 2011 HSC 2c

Find an expression for the coefficient of x^2 in the expansion of $\left(3x - \frac{4}{x}\right)^8$. ***(2 marks)***

4. Combinatorics, EXT1 A1 2020 HSC 14a

i. Use the identity $(1+x)^{2n} = (1+x)^n(1+x)^n$

to show that

$$\binom{2n}{n} = \binom{n}{0}^2 + \binom{n}{1}^2 + \ldots + \binom{n}{n}^2,$$

where n is a positive integer. ***(2 marks)***

ii. A club has $2n$ members, with n women and n men.

A group consisting of an even number $(0, 2, 4, \ldots, 2n)$ of members is chosen, with the number of men equal to the number of women.

Show, giving reasons, that the number of ways to do this is $\binom{2n}{n}$. ***(2 marks)***

iii. From the group chosen in part (ii), one of the men and one of the women are selected as leaders.

Show, giving reasons, that the number of ways to choose the even number of people and then the leaders is

$$1^2\binom{n}{1}^2 + 2^2\binom{n}{2}^2 + \ldots + n^2\binom{n}{n}^2.$$ ***(2 marks)***

iv. The process is now reversed so that the leaders, one man and one woman, are chosen first. The rest of the group is then selected, still made up of an equal number of women and men.

By considering this reversed process and using part (ii), find a simple expression for the sum in part (iii). ***(2 marks)***

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Worked Solutions

1. Combinatorics, EXT1 A1 2014 HSC 3 MC

General term:

$$T_k = \binom{12}{k}(2x)^{12-k} \cdot (-1)^k \cdot \left(5x^{-3}\right)^k$$

$$= \binom{12}{k} 2^{12-k} \cdot x^{12-k} \cdot (-1)^k \cdot 5^k \cdot x^{-3k}$$

$$= \binom{12}{k}(-1)^k \cdot 2^{12-k} \cdot 5^k \cdot x^{12-4k}$$

Constant term occurs when:

$12 - 4k = 0 \quad \Rightarrow \quad k = 3$

$\therefore$ Constant term $= \binom{12}{3}(-1)^3 \cdot 2^9 \cdot 5^3 = -\binom{12}{3} \cdot 2^9 \cdot 5^3$

$\Rightarrow C$

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2. Combinatorics, EXT1′ A1 2016 HSC 6 MC

$p(x+1) = 1 + (x+1) + (x+1)^2 + \ldots + (x+1)^{12}$

Coefficient of x^8

$= {}^{12}C_8 + {}^{11}C_8 + {}^{10}C_8 + {}^{9}C_8 + {}^{8}C_8$

$= 715$

$\Rightarrow C$

Worked Solutions

3. Combinatorics, EXT1 A1 2011 HSC 2c

$$\left(3x - \frac{4}{x}\right)^8 = \sum_{k=0}^{8} {}^8C_k \cdot (3x)^{8-k} \cdot \left(-\frac{4}{x}\right)^k$$

$$\begin{aligned}\text{General term} &= {}^8C_k \cdot 3^{8-k} \cdot x^{8-k} \cdot (-1)^k \cdot 4^k \cdot x^{-k} \\ &= {}^8C_k \cdot (-1)^k \cdot 3^{8-k} \cdot 4^k \cdot x^{8-2k}\end{aligned}$$

Co-efficient of $x^2 = 2$ occurs when:

$8 - 2k = 2 \quad \Rightarrow \quad k = 3$

$\therefore$ Co-efficient of x^2

$$\begin{aligned}&= {}^8C_3 \cdot (-1)^3 \cdot 3^5 \cdot 4^3 \\ &= -56 \times 243 \times 64 \\ &= -870\ 912\end{aligned}$$

4. Combinatorics, EXT1 A1 2020 HSC 14a

i. Expand $(1+x)^{2n}$:

$${}^{2n}C_0 + {}^{2n}C_1 x^2 + \ldots + {}^{2n}C_n x^n + \ldots {}^{2n}C_{2n} x^{2n}$$

$\Rightarrow$ Coefficient of $x^n = {}^{2n}C_n$

♦♦ *Mean mark part (i) 26%.*

Expand $(1+x)^n(1+x)^n$:

$$[{}^nC_0 + {}^nC_1 x + \ldots + {}^nC_n x^n][{}^nC_0 + {}^nC_1 x + \ldots + {}^nC_n x^n]$$

$\Rightarrow$ Coefficient of x^n

$$= {}^nC_0 \cdot {}^nC_n + {}^nC_1 \cdot {}^nC_{n-1} + \ldots + {}^nC_{n-1} \cdot {}^nC_1 + {}^nC_n \cdot {}^nC_0$$

$$= ({}^nC_0)^2 + ({}^nC_1)^2 + \ldots + ({}^nC_{n-1})^2 + ({}^nC_n)^2 \quad ({}^nC_k = {}^nC_{n-k})$$

Equating coefficients:

$${}^{2n}C_n = ({}^nC_0)^2 + ({}^nC_1)^2 + \ldots + ({}^nC_n)^2$$

♦♦ *Mean mark part (ii) 23%.*

ii. Number of men = Number of women $(M = W)$

If $M = W = 0$: Ways $= {}^nC_0 \cdot {}^nC_0 = ({}^nC_0)^2$

If $M = W = 1$: Ways $= {}^nC_1 \cdot {}^nC_1 = ({}^nC_1)^2$

$\vdots$

If $M = W = n$: Ways $= {}^nC_n \cdot {}^nC_n = ({}^nC_n)^2$

$\therefore$ Total combinations

$$= ({}^nC_0)^2 + ({}^nC_1)^2 + \ldots + ({}^nC_n)^2$$

$$= {}^{2n}C_n \quad \text{(from part (i))}$$

iii. Let M_L = possible male leaders

Let W_L = possible female leaders

♦♦ *Mean mark part (iii) 26%.*

If $M = W = 0 \Rightarrow$ no leaders

If $M = W = 1$: Ways $= {}^nC_1 \times M_L \times {}^nC_1 \times W_L = 1^2({}^nC_1)^2$

If $M = W = 2$: Ways $= {}^nC_2 \times 2 \times {}^nC_2 \times 2 = 2^2({}^nC_2)^2$

$\vdots$

If $M = W = n$: Ways $= {}^{n}C_n \times n \times {}^{n}C_2 \times n = n^2({}^{n}C_n)^2$

$\therefore$ Total combinations

$= 1^2({}^{n}C_1)^2 + 2^2({}^{n}C_2)^2 + \ldots + n^2({}^{n}C_n)^2$

♦♦♦ *Mean mark part (iv) 16%.*

iv. If $M = W = 1$: Ways $= M_L \times {}^{n-1}C_0 \times W_L \times {}^{n-1}C_0$

$= n \times {}^{n-1}C_0 \times n \times {}^{n-1}C_0$

$= n^2\left({}^{n-1}C_0\right)^2$

If $M = W = 2$: Ways $= n \times {}^{n-1}C_1 \times n \times {}^{n-1}C_1$

$= n^2\left({}^{n-1}C_1\right)^2$

$\vdots$

If $M = W = n$: Ways $= n \times {}^{n-1}C_{n-1} \times n \times {}^{n-1}C_{n-1}$

$= n^2\left({}^{n-1}C_{n-1}\right)^2$

$\therefore$ Total combinations

$= n^2\left({}^{n-1}C_0\right)^2 + n^2\left({}^{n-1}C_1\right)^2 + \ldots + n^2\left({}^{n-1}C_{n-1}\right)^2$

$= n^2 \times {}^{2(n-1)}C_{n-1}$ (using part (i))

$= n^2 \times {}^{2n-2}C_{n-1}$

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